

Surjective Homomorphisms from the Plactic Monoid: the Optimal Exponent Is n . A Proof of Conjecture 00000003841

GitHub: gaochengzhecpu

Abstract

Let P_n be the plactic monoid on $n \geq 1$ letters. We prove that a real number c has the property that every finite monoid M receives at most $|M|^c$ surjective homomorphisms from P_n if and only if $c \geq n$. So a constant $c(n)$ as in the source exists, the optimal one is $c(n) = n$, it is at least 1, and it is unbounded in n . The statement and its proof are formalised in Lean with Mathlib, including the plactic monoid and real exponents.

Statement and reading of the source

Let $A_n = \{1 < 2 < \dots < n\}$ and let A_n^* be the free monoid of words. Knuth equivalence is the monoid congruence generated by

$$acb \equiv cab \quad (a \leq b < c), \quad bac \equiv bca \quad (a < b \leq c),$$

and the plactic monoid is $P_n = A_n^*/\equiv$. For a monoid M let $\text{Sur}(P_n, M)$ be the set of surjective monoid homomorphisms $P_n \rightarrow M$. Call a real number c *admissible for n* if

$$|\text{Sur}(P_n, M)| \leq |M|^c \quad \text{for every finite monoid } M.$$

The source, in both languages, asserts three things: an admissible constant $c(n)$ exists; the optimal exponent satisfies $c(n) \geq 1$; and the optimal exponent grows unboundedly with n . We read “optimal exponent” as the least admissible real number. The source does not say that $c(n)$ is an integer, so real exponents are allowed.

Theorem 1. *Let $n \geq 1$. A real number c is admissible for n if and only if $c \geq n$.*

Corollary 2. *For every $n \geq 1$ the least admissible exponent exists and equals n . In particular $c(n) \geq 1$, and $c(n) \rightarrow \infty$. The conjecture holds.*

Proof

Lemma 3. *For every $n \geq 0$ and every finite monoid M , the number of all homomorphisms $P_n \rightarrow M$ is at most $|M|^n$.*

Proof. The classes $[1], \dots, [n]$ of the letters generate P_n , because every word is a product of letters. So a homomorphism $f: P_n \rightarrow M$ is determined by the tuple $(f[1], \dots, f[n]) \in M^n$. $\square$

Lemma 4. *Let M be a commutative monoid. Every map $g: A_n \rightarrow M$ extends to a homomorphism $\bar{g}: P_n \rightarrow M$ with $\bar{g}[i] = g(i)$.*

Proof. The map g extends to a homomorphism $A_n^* \rightarrow M$. In a commutative monoid the two sides of each Knuth relation have the same image, since they are products of the same three elements. So the extension is constant on Knuth classes. $\square$

Lemma 5. *Let p be a prime and $n \geq 0$. Then $|\text{Sur}(P_n, \mathbb{Z}/p)| \geq p^n - 1$, where $\mathbb{Z}/p$ is the cyclic group of order p regarded as a monoid.*

Proof. Let $g: A_n \rightarrow \mathbb{Z}/p$ be a map that is not identically 0, say $g(i) = a \neq 0$. Let $b \in \mathbb{Z}/p$, and let $k \in \{0, \dots, p-1\}$ represent ba^{-1} . Then $\bar{g}([i]^k) = ka = b$. So $\bar{g}$ is surjective. Different maps g give different homomorphisms, and there are $p^n - 1$ maps that are not identically 0. (In fact equality holds, since the homomorphism with $g = 0$ is not surjective.) $\square$

Proof of Theorem 1. Let $c \geq n$ and let M be a finite monoid. Then $|M| \geq 1$, so by Lemma 3 $|\text{Sur}(P_n, M)| \leq |M|^n \leq |M|^c$. So c is admissible.

Let $c < n$ and suppose that c is admissible. Put $\delta = n - c > 0$ and choose a prime $p \geq 3$ with $p \geq 2^{1/\delta}$. Then $p^\delta \geq 2$, so $p^c = p^n/p^\delta \leq p^n/2$. Admissibility for $M = \mathbb{Z}/p$ and Lemma 5 give

$$p^n - 1 \leq |\text{Sur}(P_n, \mathbb{Z}/p)| \leq p^c \leq p^n/2,$$

hence $p^n \leq 2$. But $p^n \geq p \geq 3$ because $n \geq 1$. This contradiction shows that c is not admissible. $\square$

Proof of Corollary 2. By Theorem 1 the set of admissible exponents for $n \geq 1$ is $[n, \infty)$. Its least element is $n \geq 1$, and for every real bound B all admissible exponents for all $n \geq \max(B, 1)$ are at least B . $\square$

Remark. (1) *The case $n = 0$.* P_0 is the trivial monoid. It has one surjective homomorphism to a trivial monoid and none to any other, so every real number is admissible and there is no least one. The source speaks of the alphabet $\{1, \dots, n\}$, and the statement about the optimal exponent is proved for $n \geq 1$.

(2) *What is used about P_n .* The proof uses two properties: P_n is generated by n elements, and the Knuth relations hold in every commutative monoid. The same proof therefore gives the same answer for the free monoid, the free commutative monoid and the hypoplactic monoid on n letters. The result is elementary; the fine structure of the plactic monoid does not enter.

(3) *Counting quotients.* The label of the source is “plactic quotient count bound”, while its statement in both languages counts surjective homomorphisms, and this is what is proved here. If one counted surjective homomorphisms up to automorphisms of M , that is, congruences on P_n with quotient isomorphic to M , then $|M|^n$ is still a bound and $\mathbb{Z}/p$ has $(p^n - 1)/(p - 1) \geq p^{n-1}$ such congruences, so the optimal exponent would lie in $[n - 1, n]$ and would still be unbounded; but for $n = 1$ it would be 0, so the clause $c(n) \geq 1$ would hold only from $n = 2$ on. This variant is not formalised.

Formal verification and scope

`KnuthRel` is the pair of Knuth relations on Mathlib’s `FreeMonoid` of a linearly ordered alphabet, `placticCon` is the congruence generated by it (Mathlib’s `conGen`), and `Plactic n` is the quotient monoid for the alphabet `Fin n`. Homomorphisms are Mathlib’s monoid homomorphisms, and

$$\text{numSurj } n \ M$$

is the cardinality of the type of surjective homomorphisms from `Plactic n` to M . Lemma 3 is `hom_ext_gen` and `numSurj_le`. Lemma 4 is the definition `liftComm` with `liftComm_gen`. Lemma 5 is

$$\text{surjective_liftComm, } \text{pow_le_numSurj_add_one,}$$

for the group `Multiplicative (ZMod p)`.

`IsAdmissible n c` is the displayed inequality for all finite monoids, with c a real number and Mathlib’s real power. Theorem 1 is `isAdmissible_iff`, and the least-element statement is

`isLeast_admissible`. The proposition **Conjecture** is the conjunction of: for every n there is an admissible exponent; for every $n \geq 1$ there is a least admissible exponent and it is at least 1; for every real B there is N such that for all $n \geq N$ every admissible exponent is at least B . The final theorem is

`conjecture_true : Conjecture.`

No sorry, `native_decide` or custom axiom occurs.

Everything in Theorem 1 and Corollary 2 is formalised. The quantification is over monoids in the lowest universe, which contains a copy of every finite monoid. The three points of the remark are not formalised.

Source and reproduction

The exact bilingual source is included byte-for-byte as `SOURCE.md`. Its repository path is

`conjectures/00000003841.md`.

The project pins Lean 4.19.0 and Mathlib commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

From `lean/`, run `lake build`, then

`lake env lean -DwarningAsError=true Main.lean`.

The included public-Git manifest locks all dependency revisions. The `verification/` directory records the fresh-build logs, theorem-axiom reports, artifact hashes, review and final PDF inspection. No supplementary numerical computation is required.